Daniel R. Palomo

St. Louis, MO | d.palomo@wustl.edu | (713) 702-7657

EDUCATION

Washington University in St. Louis *St. Louis, MO*
Major: Biomedical Engineering Expected May 2028
Minor: Biomedical Data Science
Cumulative & Program GPA: 3.82/4.00

Relevant Coursework: Data Structures and Algorithms, Introduction to Computer Science, Engineering Math B, Differential Equations, Biomechanics, Physiological Control Systems

RESEARCH EXPERIENCE

Scheller Lab | Washington University School of Medicine *St. Louis, MO*
Undergraduate Research Assistant August 2025 - Present

- Analyzed micro-computed tomography (microCT) imaging scans to assess cortical bone parameters, including cortical thickness and bone volume, across varying molecular treatment and mechanical loading conditions.
- Developed Python-based data analysis and visualization workflows to generate comparative plots and summarize treatment effects across experimental groups.
- Validated microCT measurements by comparing commercial and open-source analysis platforms across various segmentation thresholds and implementing normalization strategies to samples with variable scan ranges.
- Executed 3D volumetric image processing and spatial registration in Dragonfly prior to structural analysis.
- Authored two 3,600-word independent reports with 8+ figures detailing analysis methods, validation, and results.

Biomedical Engineering and Informatics REU | Wake Forest University School of Medicine *Winston-Salem, NC*
Undergraduate Research Assistant May 2026 - July 2026

- Selected for NIH-funded 10-week research program investigating weight-loss associated bone loss.
- Will analyze CT and HRpQCT imaging data to quantify bone mineral density, cortical structure, and estimated mechanical strength.
- Contribute to finite element modeling and hypothesis-driven evaluation of exercise and pharmacologic interventions.

PUBLICATIONS AND MANUSCRIPTS

- Meslier, Q. A., Beeve, A. T., Gupta, A., **Palomo, D.,** Saleem, S., Eck, S., Lawson, L., Shuster, J., Brennan, M., Dirckx, N., Silva, M. J., & Scheller, E. L. (2025). *Spatial transcriptomics for gene discovery identifies Slc13a5 as a modulator of bone mechanoadaptation.*
  - Manuscript under review at the Journal of Bone and Mineral Research.

PRESENTATIONS

- **Palomo, D.,** Meslier, Q. A., Dirckx, N., Scheller, E. L.
  Role of Slc13a5 in Osteoblast-Mediated Bone Remodeling and Marrow Fat Regulation.
  - Poster Presentation, Washington University Undergraduate Research Symposium, Spring 2026

SKILLS

- *Programming Languages:* Python, R, Java, AutoCAD
- *Computational Skills:* Finite Element Modeling, ImageJ, Dragonfly, ScanCo Evaluation
- *Laboratory Techniques:* MicroCT Imaging, Gel Electrophoresis, Cell Culturing, Microscopy, Pipetting
- *Soft Skills*: Fluency in Spanish, Team Management, Conflict Resolution

Daniel R. Palomo

St. Louis, MO | d.palomo@wustl.edu | (713) 702-7657

TEACHING EXPERIENCE

Introduction to Biomedical Engineering | BME 1400 — *St. Louis, MO*
Teaching Assistant — August 2025 - December 2025

- Graded exams, weekly assignments, and lab reports for 100+ students.
- Facilitated laboratory sessions in biomechanics, electrophysiology, and core BME concepts.
- Held weekly office hours via Zoom to reinforce course material and support student learning.

LEADERSHIP AND SERVICE

Washington University Biomedical Engineering Society (BMES) — *St. Louis, MO*
President — August 2025 - Present

- Lead organizational operations and delegate responsibilities across the executive board.
- Manage budgeting, event logistics, and space reservations for organization-wide initiatives.
- Coordinate recruitment and outreach with biomedical companies and academic programs.
- Represent BMES in engineering courses and departmental events to promote student engagement.

Professional Development Representative — October 2024 - August 2025

- Organized professional events in conjunction with upperclassman representatives, including information sessions with leading biomedical companies and career panels featuring professionals, academics, and medical students.

Bear Cubs Running Team — *St. Louis, MO*
Coach & Mentor — September 2024 - Present

- Mentor youth with special needs through structured running activities to support motor development, social skills, and confidence.
- Foster inclusive team environments emphasizing teamwork, consistency, and physical activity.

Beta Theta Pi | Alpha Iota Chapter — *St. Louis, MO*
Member — August 2025 - Present

- Contributed to fundraising operations that collected over $2000 for the HardyStrong Foundation, a 501(c)(3) dedicated to furthering research for gastric cancer.

ATHLETICS

NCAA Track and Field — *St. Louis, MO*
Varsity Athlete — August 2024 - Present

- Compete in indoor and outdoor NCAA seasons.
- Commit 20+ hours/week to training, competition, and team responsibilities.

AWARDS AND HONORS

- Dean’s List (4x) - Fall 2024; Spring 2025; Fall 2025; Spring 2026